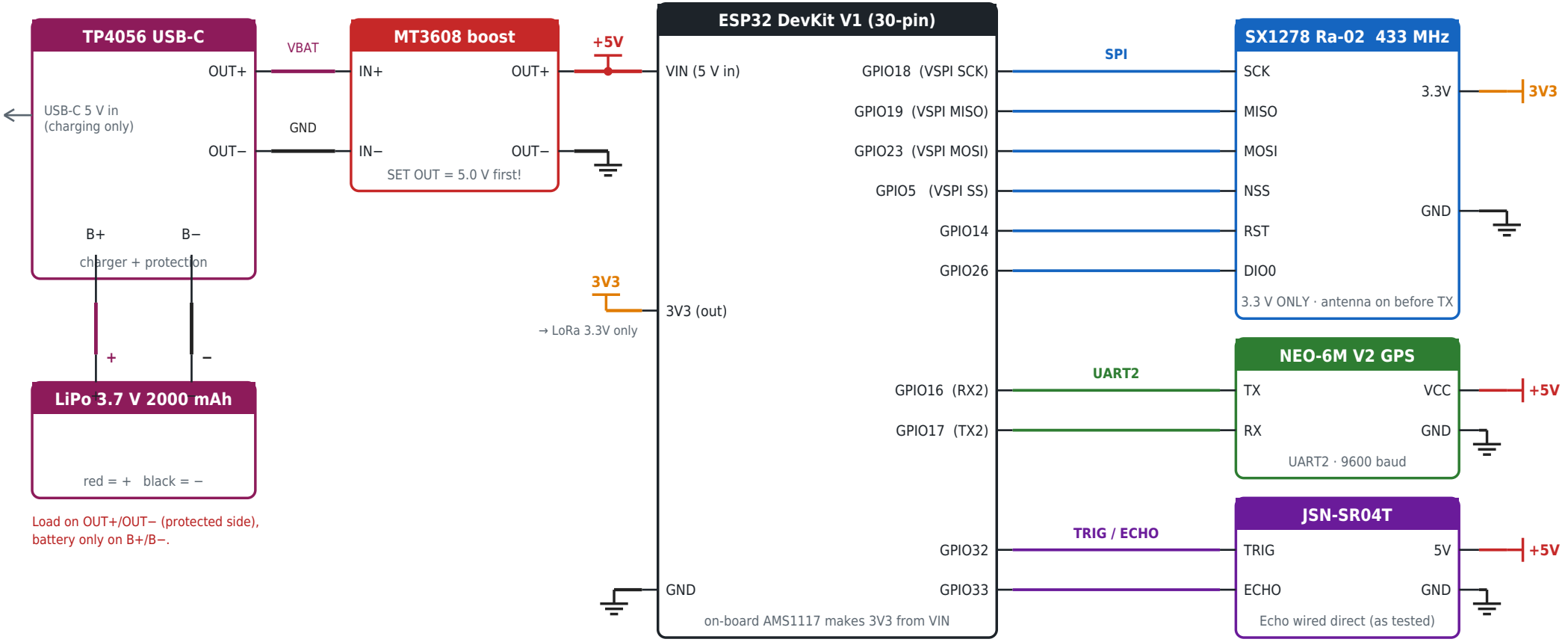
ESP32 LoRa Flood Sensor — Wiring Guide (Prototype v1)

Schematic — every connection in the build. Colours = nets (see legend).

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Power path

LiPo → **TP4056** (charges via USB-C, protects the cell) → **MT3608** boosts 3.0–4.2 V up to **5.0 V** → **+5V rail**: ESP32 VIN, GPS VCC, JSN-SR04T 5V.
ESP32's on-board regulator outputs **3V3**, which feeds only the Ra-02 LoRa module.
All GNDs are one common net.

Before the first power-up

1. Connect battery to TP4056, MT3608 IN to TP4056 OUT, **nothing** on MT3608 OUT.
2. Turn the MT3608 trimmer while measuring OUT with a multimeter until it reads **5.0 V** (it can ship set much higher).
3. Screw the 433 MHz antenna onto the Ra-02 — transmitting without it can damage the radio.
4. Never feed 5 V into the Ra-02.

Net colours

- +5V rail (MT3608 OUT+)
- 3V3 (ESP32 regulator)
- Battery 3.0–4.2 V
- GND (common)
- LoRa SPI / control
- GPS UART2
- JSN-SR04T trigger/echo

Parts: ESP32 DevKit V1 (30-pin) · SX1278 Ra-02 433 MHz · NEO-6M V2 GPS · JSN-SR04T · 3.7 V 2000 mAh LiPo · TP4056 USB-C · MT3608 boost

Match pins by the silkscreen label printed on each board, not by position.

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Pin-by-pin connection table — use this as the master reference when soldering and coding.

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Module	Module pin	Connects to	Net	Notes
LiPo battery	+ (red)	**TP4056 B+**	**VBAT**	Check polarity before plugging in.
LiPo battery	− (black)	**TP4056 B−**	**BAT−**	Battery side only — not system GND.
TP4056	OUT+	**MT3608 IN+**	**VBAT**	3.0–4.2 V, protected output.
TP4056	OUT−	**MT3608 IN−**	**GND**	This is the system ground.
MT3608	OUT+	**ESP32 VIN · GPS VCC · JSN 5V**	**+5V**	Adjust to 5.0 V before connecting (perfboard J4-1).
MT3608	OUT−	**ESP32 GND · all module GNDs**	**GND**	Perfboard J4-2.
SX1278 Ra-02	3.3V	**ESP32 3V3**	**3V3**	3.3 V only — 5 V will destroy it.
SX1278 Ra-02	GND	**GND**	**GND**	
SX1278 Ra-02	SCK	**GPIO18**	**SPI**	VSPI default clock.
SX1278 Ra-02	MISO	**GPIO19**	**SPI**	VSPI default MISO.
SX1278 Ra-02	MOSI	**GPIO23**	**SPI**	VSPI default MOSI.
SX1278 Ra-02	NSS	**GPIO5**	**SPI**	Chip select.
SX1278 Ra-02	RST	**GPIO14**	**LoRa**	Reset.
SX1278 Ra-02	DIO0	**GPIO26**	**LoRa**	TX-done / RX-done interrupt.
NEO-6M V2	VCC	**+5V rail**	**+5V**	Board has its own 3.3 V regulator (3V3 also works if that is what you tested).
NEO-6M V2	GND	**GND**	**GND**	
NEO-6M V2	TX	**GPIO16 (RX2)**	**UART2**	GPS talks → ESP32 listens. 9600 baud.
NEO-6M V2	RX	**GPIO17 (TX2)**	**UART2**	Optional for reading; needed only to configure the GPS.
JSN-SR04T	5V	**+5V rail**	**+5V**	
JSN-SR04T	Trig	**GPIO32**	**JSN**	10 µs HIGH pulse starts a reading.
JSN-SR04T	Echo	**GPIO33**	**JSN**	Direct to ESP32, as you tested.
JSN-SR04T	GND	**GND**	**GND**	

Why these GPIOs

- **18 / 19 / 23 / 5** are the ESP32's default VSPI pins, so `SPI.begin()` works without remapping.
- **16 / 17** are the default UART2 (Serial2) pins.
- **32 / 33** are plain I/O pins — safe for TRIG/ECHO.
- Avoided: boot-strapping pins **0, 2, 12, 15** and input-only pins **34–39**.

Already tested different pins?

Keep what works for you — just update this table, the wire list on page 3 and the `#defines` on page 4 together so the guide stays the single source of truth.

Current budget (estimate)

ESP32 active (Wi-Fi off) ≈ 50 mA · GPS ≈ 45 mA · LoRa RX ≈ 12 mA, TX bursts ≈ 120 mA · JSN ≈ 8 mA. Average ≈ 110 mA at 5 V ≈ 175 mA from the battery (≈ 85 % boost efficiency) → roughly **8–10 h** running continuously on 2000 mAh. Deep sleep between readings stretches this to days.

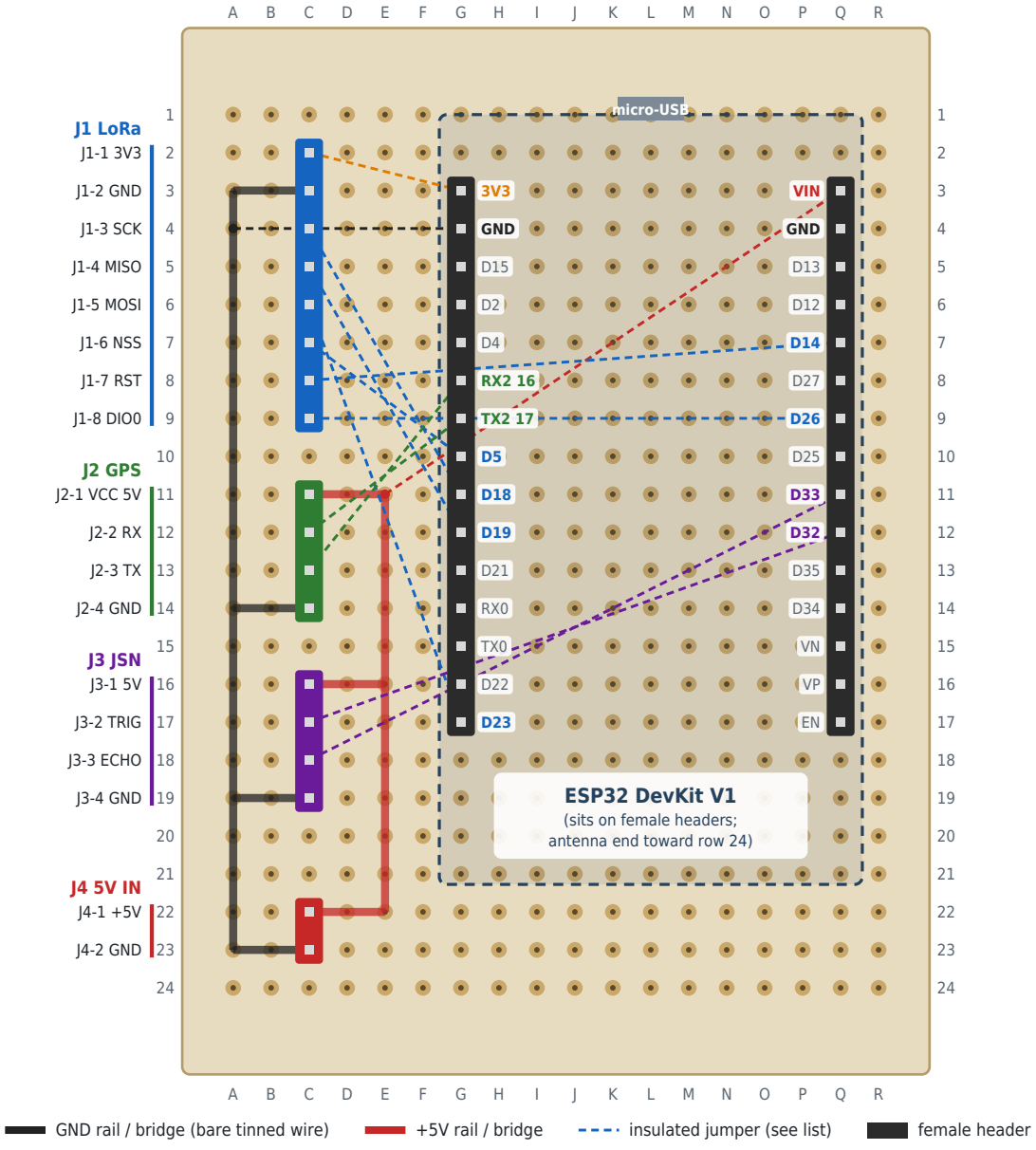
Uploading code over the ESP32 USB

Unplug the MT3608 lead from **J4** while the ESP32's own USB is connected to your computer, so the two 5 V sources never feed VIN at the same time. Plug J4 back in afterwards.

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Perfboard layout — 5 × 7 cm board (18 × 24 holes, 2.54 mm pitch), viewed from the COMPONENT side.

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#	From	To	What	Type
R1	A3	A23	GND rail (column A)	bare
R2	E11	E22	+5V rail (column E)	bare
B1	A3–C3		J1 GND → GND rail	bridge
B2	A14–C14		J2 GND → GND rail	bridge
B3	A19–C19		J3 GND → GND rail	bridge
B4	A23–C23		J4 GND → GND rail	bridge
B5	C22–E22		J4 +5V → +5V rail	bridge
B6	C16–E16		J3 5V ← +5V rail	bridge
B7	C11–E11		J2 VCC ← +5V rail	bridge
W1	E11	Q3	**+5V rail → ESP32 VIN**	jumper
W2	A4	G4	**GND rail → ESP32 GND**	jumper
W3	C2	G3	**J1-1 3V3 ← ESP32 3V3**	jumper
W4	C4	G11	**J1-3 SCK → GPIO18**	jumper
W5	C5	G12	**J1-4 MISO → GPIO19**	jumper
W6	C6	G17	**J1-5 MOSI → GPIO23**	jumper
W7	C7	G10	**J1-6 NSS → GPIO5**	jumper
W8	C8	Q7	**J1-7 RST → GPIO14**	jumper
W9	C9	Q9	**J1-8 DIO0 → GPIO26**	jumper
W10	C12	G9	**J2-2 GPS RX → GPIO17 (TX2)**	jumper
W11	C13	G8	**J2-3 GPS TX → GPIO16 (RX2)**	jumper
W12	C17	Q12	**J3-2 TRIG → GPIO32**	jumper
W13	C18	Q11	**J3-3 ECHO → GPIO33**	jumper

Read before soldering

- **Check header spacing first:** plug the ESP32 into two 15-pin female headers and drop them on the board. They should land 10 holes apart (G and Q). If yours are 9 apart, use column P instead of Q.
- **Mirror warning:** this is the top view. When you flip the board to solder, column A is on your **right**.
- Ra-02 pads are **2.0 mm** pitch — plug it into J1 via its adapter board, or solder short wires from its pads to the J1 order.
- GPS, JSN-SR04T board, TP4056 and MT3608 sit **off-board** (the ESP32 already uses most of the 5×7 cm) and plug into J2/J3/J4 with jumper wires. Fix them with tape in the enclosure.

ESP32 LoRa Flood Sensor — Wiring Guide (Prototype v1)

Build order, power-up tests, firmware pin map and mounting notes.

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Build & test order

1. **Fit check**
   Confirm the board is 18 × 24 holes and the ESP32 header spacing (page 3).
2. **ESP32 headers**
   Solder two 15-pin female headers at G3–G17 and Q3–Q17. Keeping the ESP32 plugged in while tacking the first pins keeps them straight.
3. **Connector headers**
   Solder female headers J1 (C2–C9), J2 (C11–C14), J3 (C16–C19), J4 (C22–C23).
4. **Rails & bridges**
   Run R1 (GND, column A) and R2 (+5V, column E), then bridges B1–B7.
5. **Jumpers**
   Solder insulated jumpers W1–W13. Tick each one off the list on page 3.
6. **Continuity test**
   With nothing plugged in, beep every W/B entry. Check there is NO beep between +5V–GND, 3V3–GND and 3V3–+5V.
7. **Power chain**
   Battery → TP4056 B+/B−; TP4056 OUT+/OUT− → MT3608 IN+/IN−. Adjust MT3608 OUT to 5.0 V with nothing connected.
8. **ESP32 only**
   Plug in the ESP32, connect MT3608 OUT to J4. Measure ≈ 5.0 V at VIN and ≈ 3.3 V at 3V3.
9. **Add modules one by one**
   Antenna on → plug LoRa into J1 and run a LoRa test. Then GPS on J2 (fix needs open sky, first fix can take minutes). Then JSN on J3.
10. **Full firmware**
    Load the full sketch. Remember: unplug J4 while the ESP32 USB is connected to a computer.

Firmware pin map (Arduino / ESP32 core)

```
// ===== Flood sensor pin map (matches this guide) =====
#include <SPI.h>
#include <LoRa.h>          // sandeepmistry/arduino-LoRa

#define LORA_SCK    18
#define LORA_MISO   19
#define LORA_MOSI   23
#define LORA_NSS     5
#define LORA_RST    14
#define LORA_DIO0   26
#define LORA_FREQ   433E6

#define GPS_RX      16     // ESP32 RX2  <- GPS TX
#define GPS_TX      17     // ESP32 TX2  -> GPS RX
#define GPS_BAUD  9600

#define JSN_TRIG    32
#define JSN_ECHO    33

void setup() {
  Serial.begin(115200);
  Serial2.begin(GPS_BAUD, SERIAL_8N1, GPS_RX, GPS_TX);

  pinMode(JSN_TRIG, OUTPUT);
  pinMode(JSN_ECHO, INPUT);

  SPI.begin(LORA_SCK, LORA_MISO, LORA_MOSI, LORA_NSS);
  LoRa.setPins(LORA_NSS, LORA_RST, LORA_DIO0);
  if (!LoRa.begin(LORA_FREQ)) Serial.println("LoRa init failed");
}

float readDistanceCm() {
  digitalWrite(JSN_TRIG, LOW);  delayMicroseconds(5);
  digitalWrite(JSN_TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(JSN_TRIG, LOW);
  unsigned long us = pulseIn(JSN_ECHO, HIGH, 30000);
  return us ? us * 0.0343f / 2.0f : -1;   // -1 = no echo
}
```

Turning distance into water level

level = H − d
H = distance from the sensor face down to the dry ground / channel bottom (measure once at install).
d = distance the JSN-SR04T reads.

- The JSN-SR04T can't see closer than ≈ 20–25 cm. Mount it so the highest flood you expect still stays at least 25 cm below the sensor face.
- Point it straight down with a clear cone — no walls or pipes nearby.
- Take 5 readings and use the median to ignore splashes.

Placement

- GPS antenna (ceramic patch) faces the sky, not blocked by metal.
- LoRa antenna vertical and away from the battery and the ESP32.
- Keep the TP4056 USB-C port reachable for charging.
- Keep electronics in a closed box above water; only the JSN probe goes outside.

Parts used (nothing else)

1 × ESP32 DevKit V1 (30-pin)
1 × SX1278 Ra-02 433 MHz + antenna
1 × NEO-6M V2 GPS + patch antenna
1 × JSN-SR04T (probe + board)
1 × 3.7 V 2000 mAh LiPo
1 × TP4056 USB-C charger
1 × MT3608 boost
1 × 5 × 7 cm perfboard, female headers, hookup wire